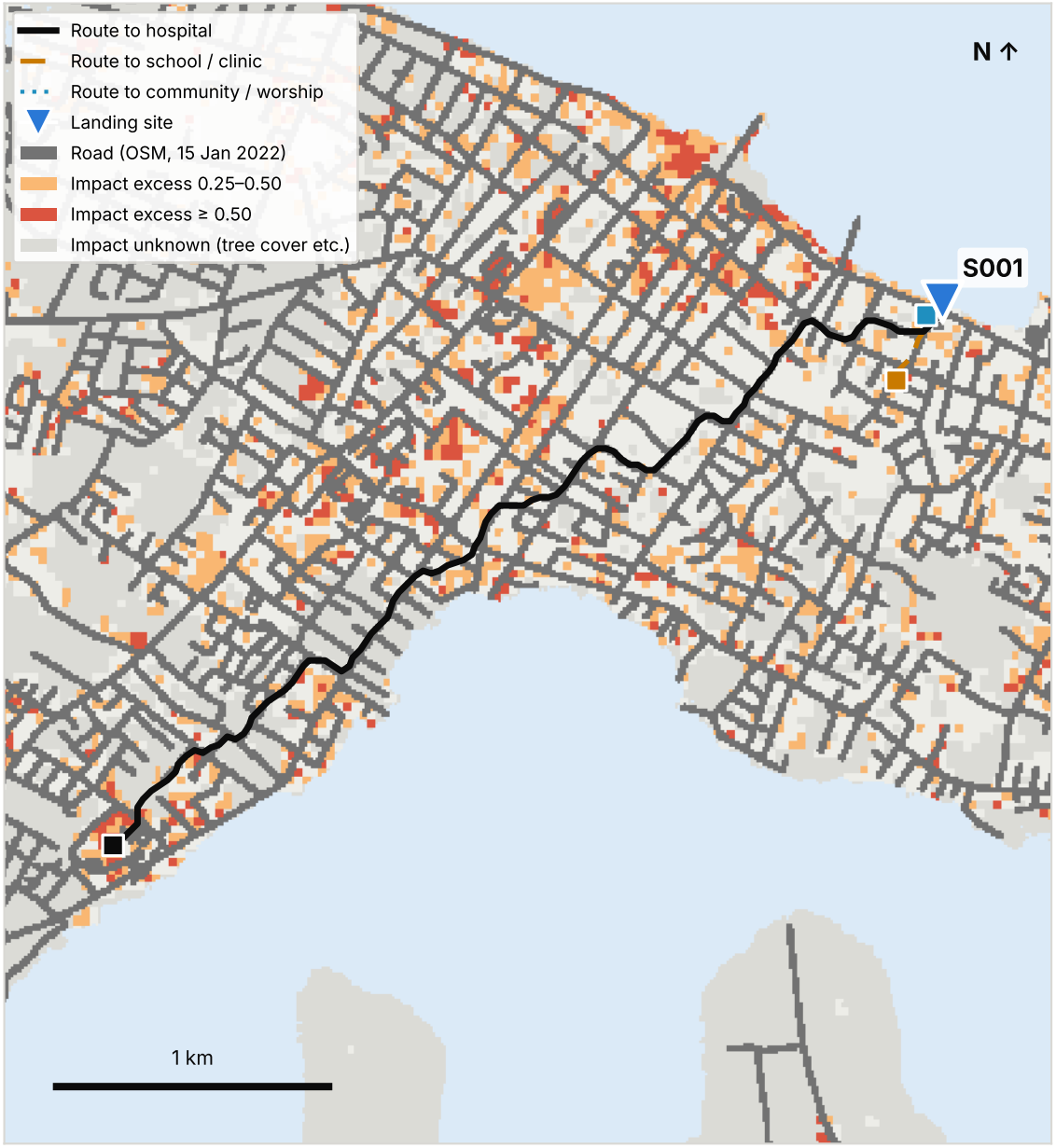
Landing site S001

Rank 1 of 112 · score 92.2 / 100 · Tongatapu, post-eruption (Jan 2022)



LOCATION

Lat / Lon	-21.13641°, -175.19027°
UTM zone 1S	E 687935 N 7661685
Lagoon mouth	2.5 km

LANDING ZONE (250 M)

Road access	28 m to road (sealed)
Impact excess	0.13 (0 = typical, 1 = max)
Impact evidence	95% of landing zone
Data confidence	**0.89**

NEAREST DESTINATION PER TIER — LEAST-COST ROUTE

Tier	Destination	km	on road
Hospital	Vaiola Hospital	**4.5**	91%
School / clinic	Fasi Clinic	0.4	90%
Community / worship	International Christian Assem...	0.1	60%

Hospital route: max slope 5°, mean impact excess 0.12, impact evidence 82% of route.

ROBUSTNESS

Rank range	**1–1 across 12 scenarios**
In top 5	12 of 12 scenarios

SCORE BREAKDOWN (POINTS)



NOT ASSESSED — VERIFY BEFORE USE

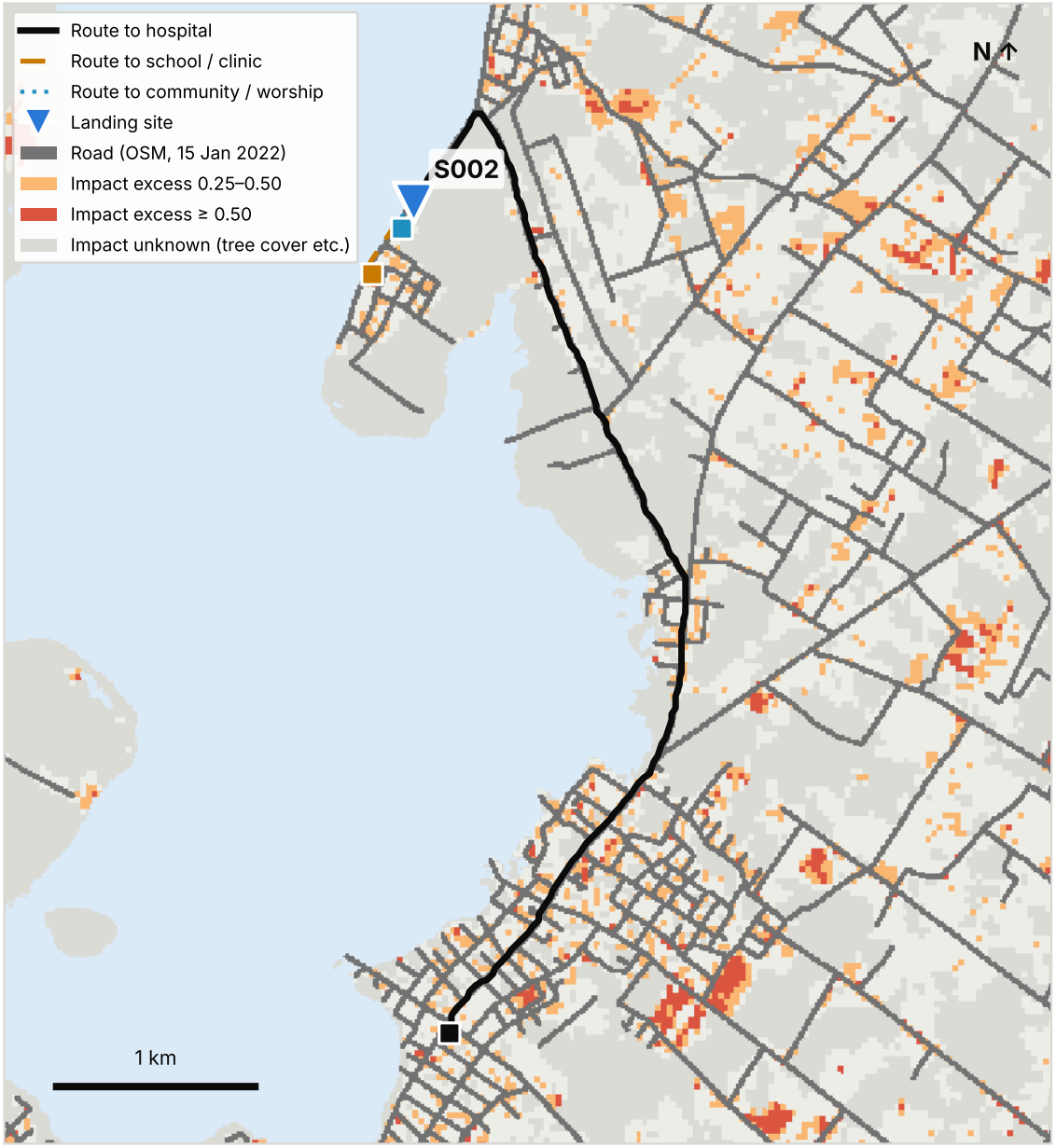
Water depth, approach and reef; beach gradient, surf and tide; obstacles; ground bearing capacity; current road and bridge condition.

Shore-Gate: Egress · screening prototype (GEOG761, University of Auckland). Data: Sentinel-1/2, ALOS-2 DPM, ESA WorldCover 2021, OSM (15 Jan 2022), Copernicus EMSR558. Score = weighted sum of five normalised criteria.
Decision support only. Results come from open satellite data and models; they must be verified by aerial reconnaissance and ground survey before operational use.

Landing site S002

Rank 2 of 112 · score 86.4 / 100 · Tongatapu, post-eruption (Jan 2022)

RECONNAISSANCE PRIORITY: HIGH
top-ranked, weak evidence; approach near lagoon mouth



LOCATION

Lat / Lon	-21.14717°, -175.12515°
UTM zone 1S	E 694685 N 7660415
Lagoon mouth	0.7 km

LANDING ZONE (250 M)

Road access	on road (sealed)
Impact excess	0.04 (0 = typical, 1 = max)
Impact evidence	13% of landing zone
Data confidence	**0.36**

NEAREST DESTINATION PER TIER — LEAST-COST ROUTE

Tier	Destination	km	on road
Hospital	Mu'a Hospital	**6.2**	99%
School / clinic	Unnamed school	0.5	97%
Community / worship	Unnamed place of worship	0.2	93%

Hospital route: max slope 13°, mean impact excess 0.11, impact evidence 59% of route.

ROBUSTNESS

Rank range	**2–4 across 12 scenarios**
In top 5	12 of 12 scenarios

SCORE BREAKDOWN (POINTS)



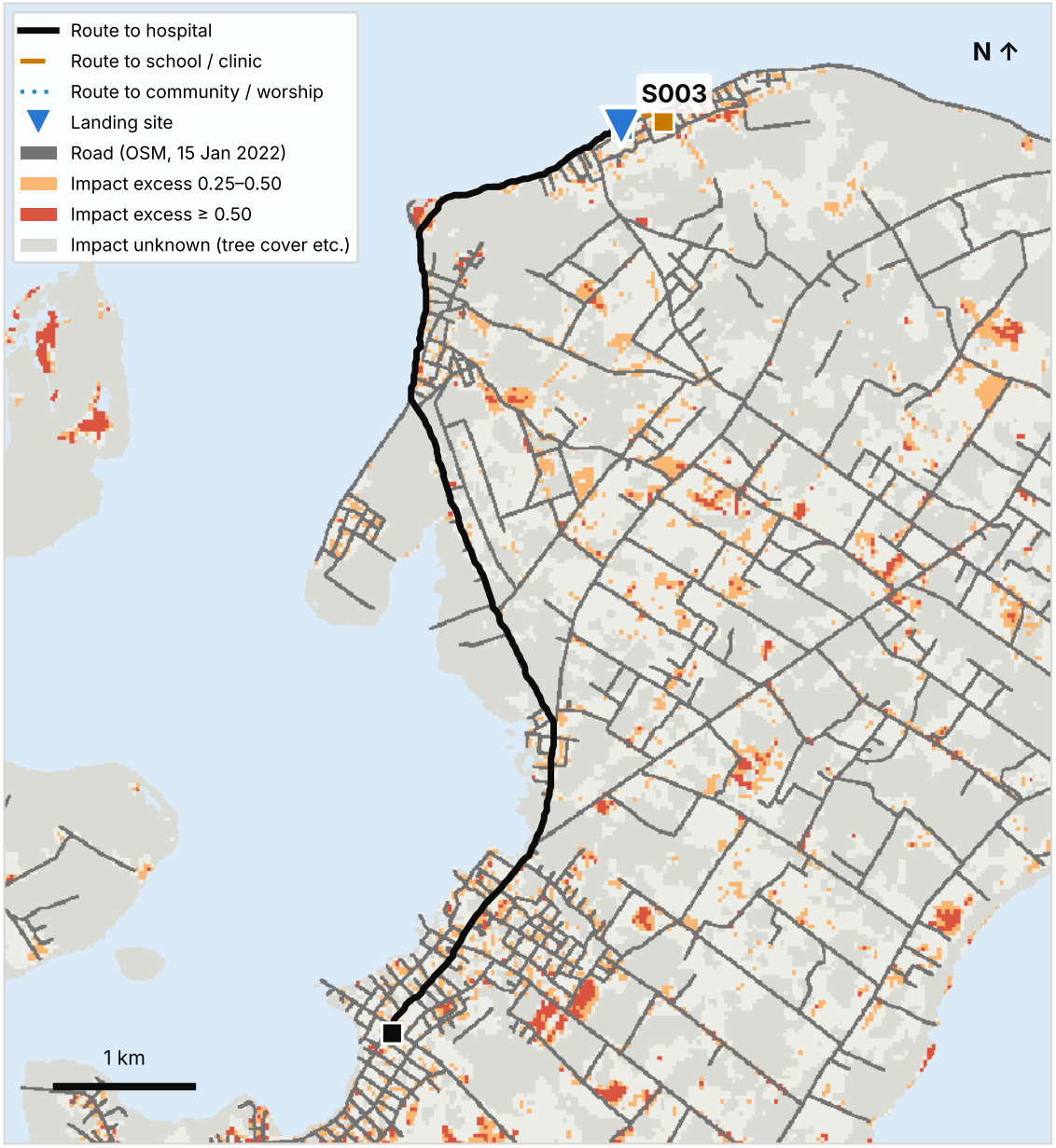
NOT ASSESSED — VERIFY BEFORE USE

Water depth, approach and reef; beach gradient, surf and tide; obstacles; ground bearing capacity; current road and bridge condition.
Approach is in the sheltered channel near the Fanga'uta lagoon entrance.
Under half of landing zone + route has impact evidence (mostly tree cover).

Shore-Gate: Egress · screening prototype (GEOG761, University of Auckland). Data: Sentinel-1/2, ALOS-2 DPM, ESA WorldCover 2021, OSM (15 Jan 2022), Copernicus EMSR558. Score = weighted sum of five normalised criteria.
Decision support only. Results come from open satellite data and models; they must be verified by aerial reconnaissance and ground survey before operational use.

Landing site S003

Rank 3 of 112 · score 86.1 / 100 · Tongatapu, post-eruption (Jan 2022)



LOCATION

Lat / Lon	-21.12448°, -175.10744°
UTM zone 1S	E 696555 N 7662905
Lagoon mouth	3.2 km

LANDING ZONE (250 M)

Road access	on road (sealed)
Impact excess	0.14 (0 = typical, 1 = max)
Impact evidence	97% of landing zone
Data confidence	**0.77**

NEAREST DESTINATION PER TIER — LEAST-COST ROUTE

Tier	Destination	km	on road
Hospital	Mu'a Hospital	**8.8**	99%
School / clinic	Unnamed school	0.4	81%
Community / worship	Unnamed community centre	0.0	67%

Hospital route: max slope 13°, mean impact excess 0.12, impact evidence 57% of route.

ROBUSTNESS

Rank range	**2–4 across 12 scenarios**
In top 5	12 of 12 scenarios

SCORE BREAKDOWN (POINTS)



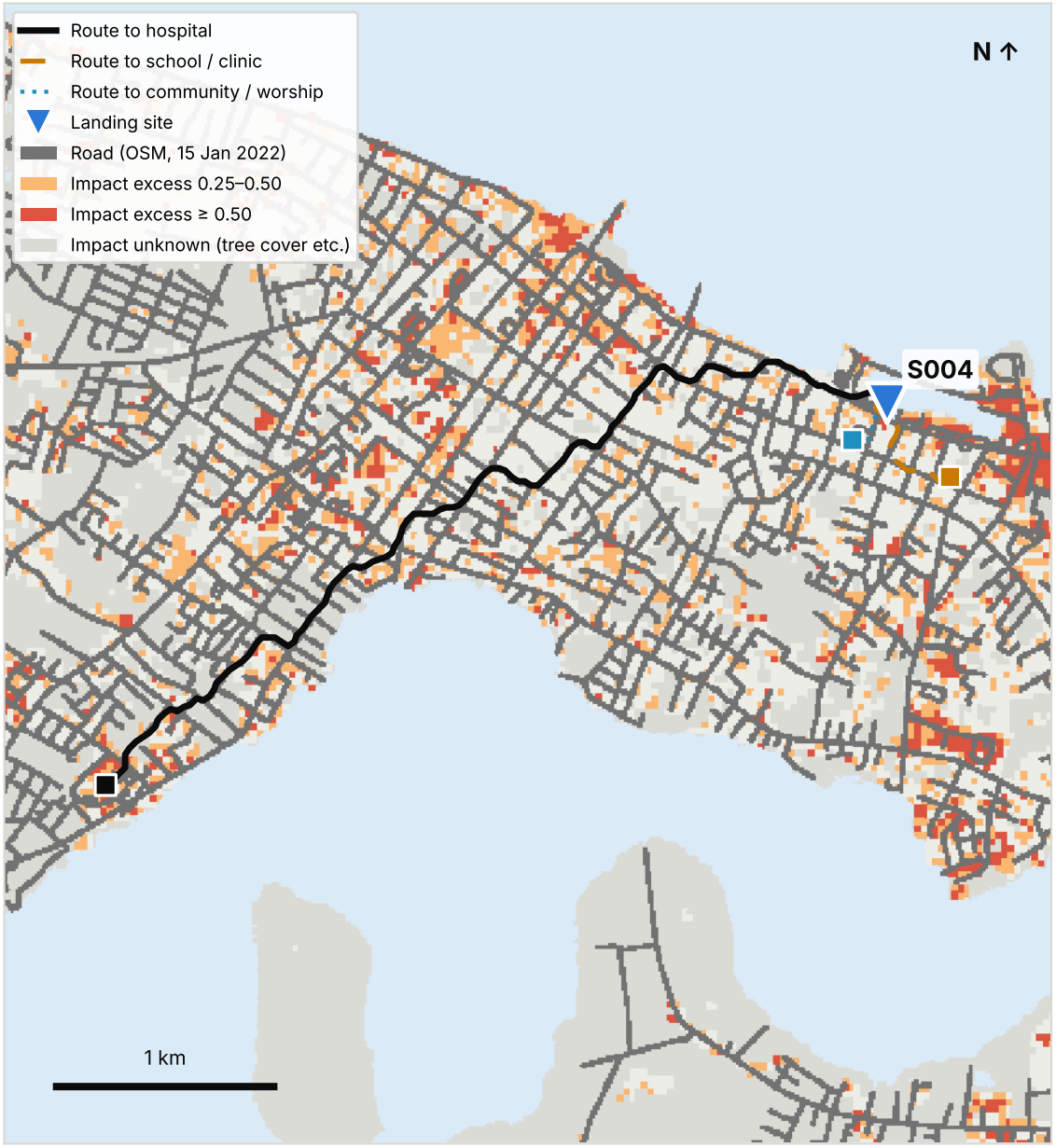
NOT ASSESSED — VERIFY BEFORE USE

Water depth, approach and reef; beach gradient, surf and tide; obstacles; ground bearing capacity; current road and bridge condition.

Shore-Gate: Egress · screening prototype (GEOG761, University of Auckland). Data: Sentinel-1/2, ALOS-2 DPM, ESA WorldCover 2021, OSM (15 Jan 2022), Copernicus EMSR558. Score = weighted sum of five normalised criteria.
Decision support only. Results come from open satellite data and models; they must be verified by aerial reconnaissance and ground survey before operational use.

Landing site S004

Rank 4 of 112 · score 85.6 / 100 · Tongatapu, post-eruption (Jan 2022)



LOCATION

Lat / Lon	-21.13852°, -175.18495°
UTM zone 1S	E 688485 N 7661445
Lagoon mouth	1.9 km

LANDING ZONE (250 M)

Road access	54 m to road (sealed)
Impact excess	0.22 (0 = typical, 1 = max)
Impact evidence	97% of landing zone
Data confidence	**0.91**

NEAREST DESTINATION PER TIER — LEAST-COST ROUTE

Tier	Destination	km	on road
Hospital	Vaiola Hospital	**5.1**	90%
School / clinic	'Ahopanilolo Institute	0.7	83%
Community / worship	Unnamed place of worship	0.4	68%

Hospital route: max slope 5°, mean impact excess 0.13, impact evidence 85% of route.

ROBUSTNESS

Rank range	**2–5 across 12 scenarios**
In top 5	12 of 12 scenarios

SCORE BREAKDOWN (POINTS)



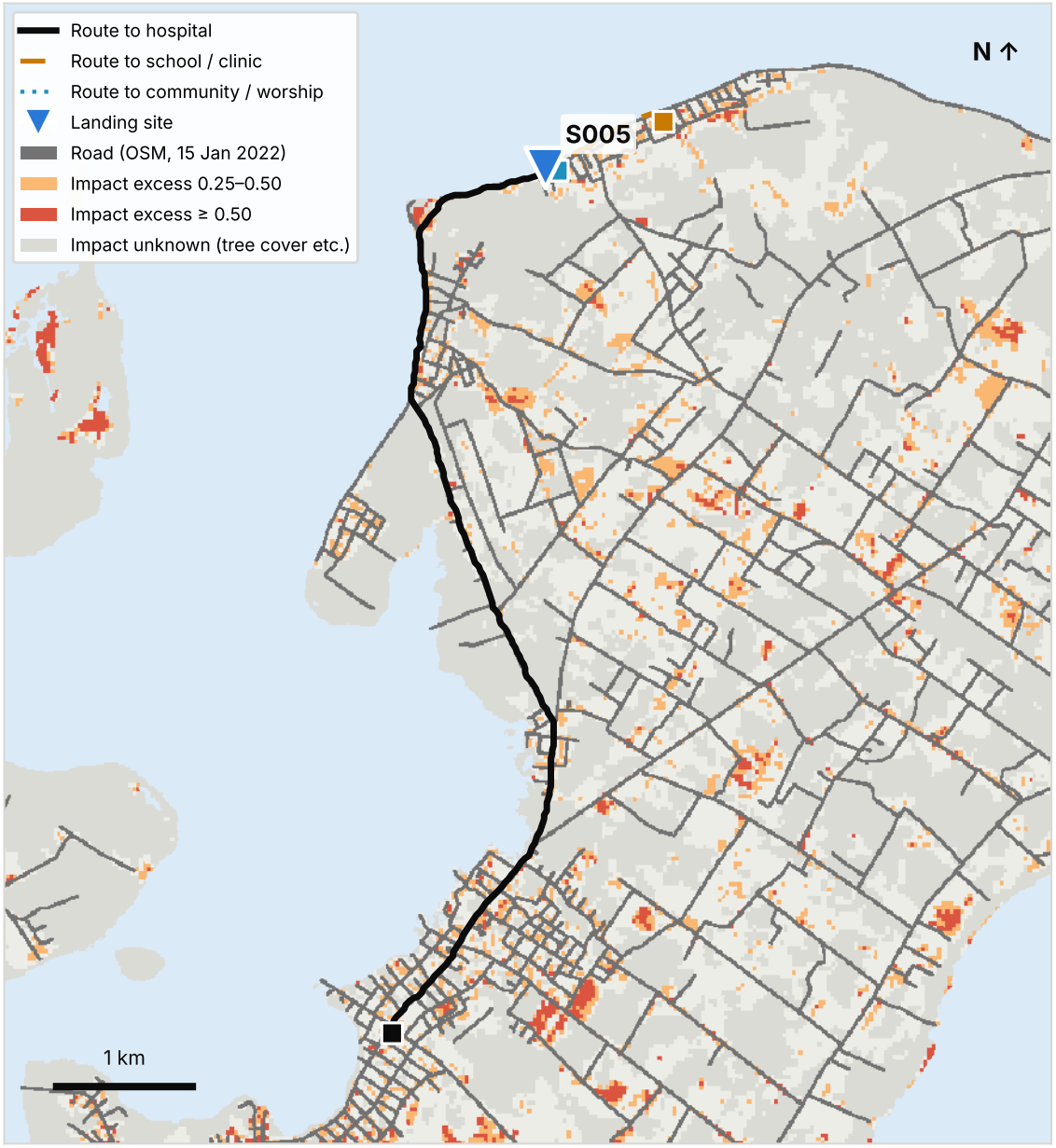
NOT ASSESSED — VERIFY BEFORE USE

Water depth, approach and reef; beach gradient, surf and tide; obstacles; ground bearing capacity; current road and bridge condition.

Shore-Gate: Egress · screening prototype (GEOG761, University of Auckland). Data: Sentinel-1/2, ALOS-2 DPM, ESA WorldCover 2021, OSM (15 Jan 2022), Copernicus EMSR558. Score = weighted sum of five normalised criteria.
Decision support only. Results come from open satellite data and models; they must be verified by aerial reconnaissance and ground survey before operational use.

Landing site S005

Rank 5 of 112 · score 84.6 / 100 · Tongatapu, post-eruption (Jan 2022)



LOCATION

Lat / Lon	-21.12725°, -175.11279°
UTM zone 1S	E 695995 N 7662605
Lagoon mouth	2.9 km

LANDING ZONE (250 M)

Road access	20 m to road (sealed)
Impact excess	0.11 (0 = typical, 1 = max)
Impact evidence	74% of landing zone
Data confidence	**0.65**

NEAREST DESTINATION PER TIER — LEAST-COST ROUTE

Tier	Destination	km	on road
Hospital	Mu'a Hospital	**8.1**	99%
School / clinic	Unnamed school	1.1	91%
Community / worship	Unnamed place of worship	0.1	73%

Hospital route: max slope 13°, mean impact excess 0.13, impact evidence 56% of route.

ROBUSTNESS

Rank range	**4–5 across 12 scenarios**
In top 5	12 of 12 scenarios

SCORE BREAKDOWN (POINTS)



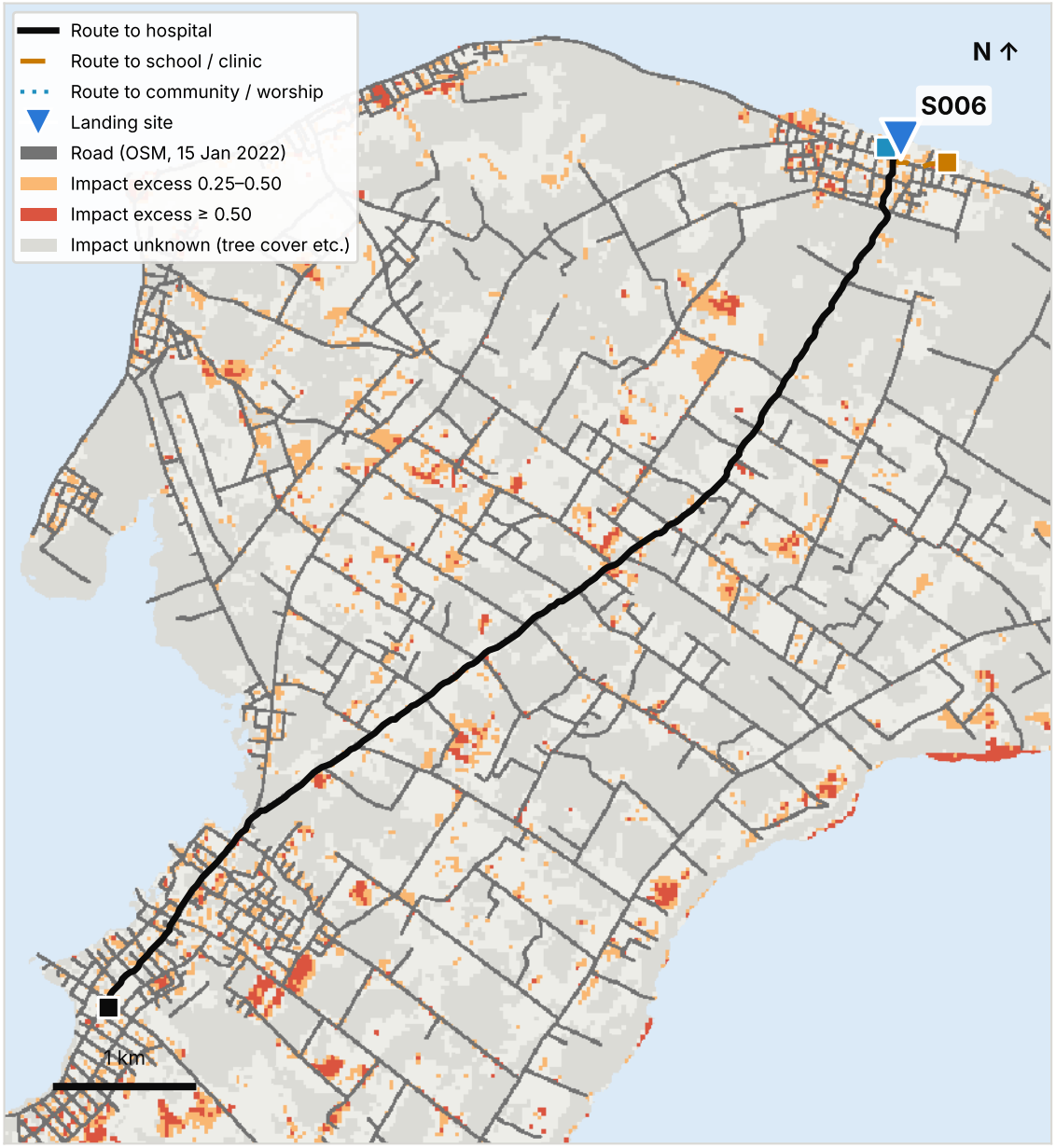
NOT ASSESSED — VERIFY BEFORE USE

Water depth, approach and reef; beach gradient, surf and tide; obstacles; ground bearing capacity; current road and bridge condition.

Shore-Gate: Egress · screening prototype (GEOG761, University of Auckland). Data: Sentinel-1/2, ALOS-2 DPM, ESA WorldCover 2021, OSM (15 Jan 2022), Copernicus EMSR558. Score = weighted sum of five normalised criteria.
Decision support only. Results come from open satellite data and models; they must be verified by aerial reconnaissance and ground survey before operational use.

Landing site S006

Rank 6 of 112 · score 83.9 / 100 · Tongatapu, post-eruption (Jan 2022)



LOCATION

Lat / Lon	-21.12665°, -175.06756°
UTM zone 1S	E 700695 N 7662615
Lagoon mouth	5.1 km

LANDING ZONE (250 M)

Road access	61 m to road (sealed)
Impact excess	0.12 (0 = typical, 1 = max)
Impact evidence	93% of landing zone
Data confidence	**0.78**

NEAREST DESTINATION PER TIER — LEAST-COST ROUTE

Tier	Destination	km	on road
Hospital	Mu'a Hospital	**9.4**	99%
School / clinic	Mo'unga 'Olive Middle School	0.6	65%
Community / worship	Unnamed place of worship	0.1	54%

Hospital route: max slope 10°, mean impact excess 0.16, impact evidence 63% of route.

ROBUSTNESS

Rank range	**6–8 across 12 scenarios**
In top 5	0 of 12 scenarios

SCORE BREAKDOWN (POINTS)



NOT ASSESSED — VERIFY BEFORE USE

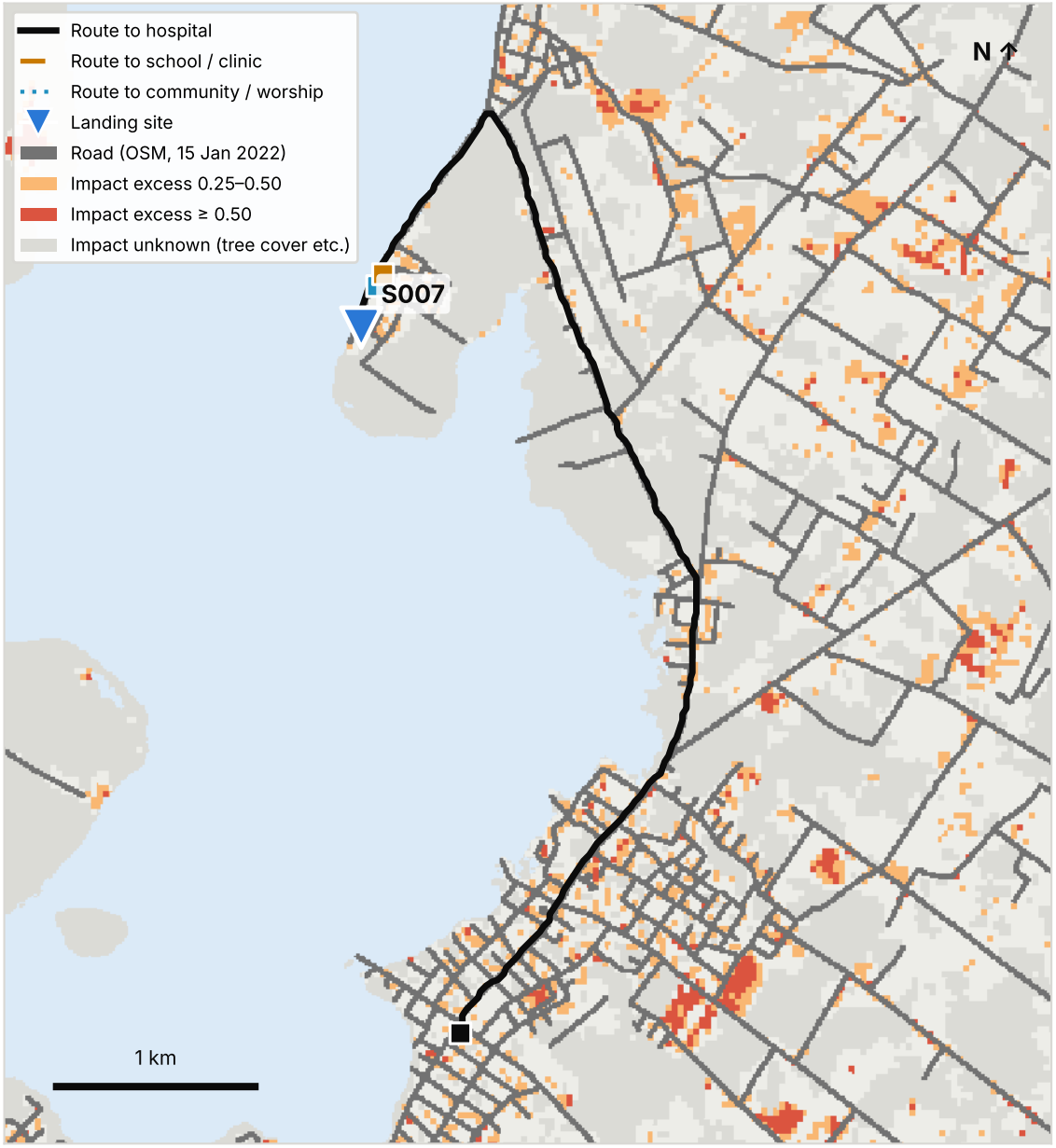
Water depth, approach and reef; beach gradient, surf and tide; obstacles; ground bearing capacity; current road and bridge condition.

Shore-Gate: Egress · screening prototype (GEOG761, University of Auckland). Data: Sentinel-1/2, ALOS-2 DPM, ESA WorldCover 2021, OSM (15 Jan 2022), Copernicus EMSR558. Score = weighted sum of five normalised criteria.
Decision support only. Results come from open satellite data and models; they must be verified by aerial reconnaissance and ground survey before operational use.

Landing site S007

Rank 7 of 112 · score 83.5 / 100 · Tongatapu, post-eruption (Jan 2022)

RECONNAISSANCE PRIORITY: HIGH
approach near lagoon mouth



LOCATION

Lat / Lon	-21.15289°, -175.12816°
UTM zone 1S	E 694365 N 7659785
Lagoon mouth	0.1 km

LANDING ZONE (250 M)

Road access	on road (sealed)
Impact excess	0.17 (0 = typical, 1 = max)
Impact evidence	55% of landing zone
Data confidence	**0.55**

NEAREST DESTINATION PER TIER — LEAST-COST ROUTE

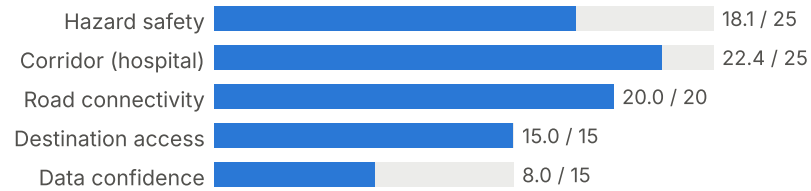
Tier	Destination	km	on road
Hospital	Mu'a Hospital	**7.0**	99%
School / clinic	Unnamed school	0.3	93%
Community / worship	Unnamed place of worship	0.3	92%

Hospital route: max slope 13°, mean impact excess 0.10, impact evidence 56% of route.

ROBUSTNESS

Rank range	**6–9 across 12 scenarios**
In top 5	0 of 12 scenarios

SCORE BREAKDOWN (POINTS)



NOT ASSESSED — VERIFY BEFORE USE

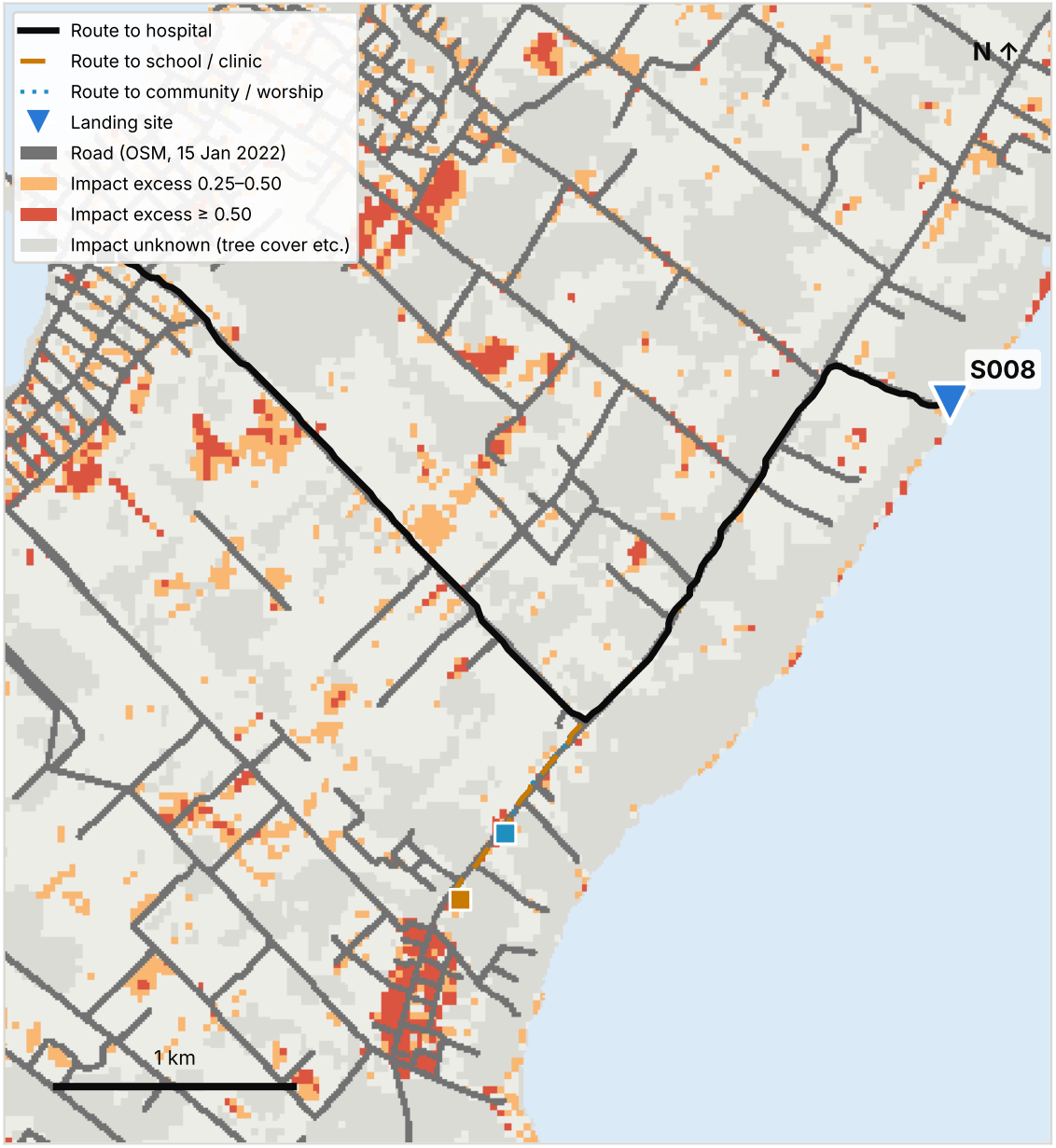
Water depth, approach and reef; beach gradient, surf and tide; obstacles; ground bearing capacity; current road and bridge condition.
Approach is in the sheltered channel near the Fanga'uta lagoon entrance.

Shore-Gate: Egress · screening prototype (GEOG761, University of Auckland). Data: Sentinel-1/2, ALOS-2 DPM, ESA WorldCover 2021, OSM (15 Jan 2022), Copernicus EMSR558. Score = weighted sum of five normalised criteria.
Decision support only. Results come from open satellite data and models; they must be verified by aerial reconnaissance and ground survey before operational use.

Landing site S008

Rank 8 of 112 · score 82.5 / 100 · Tongatapu, post-eruption (Jan 2022)

RECONNAISSANCE PRIORITY: HIGH
top-ranked, weak evidence



LOCATION

Lat / Lon	-21.19029°, -175.08849°
UTM zone 1S	E 698435 N 7655595
Lagoon mouth	4.5 km

LANDING ZONE (250 M)

Road access	51 m to road (unsealed)
Impact excess	0.10 (0 = typical, 1 = max)
Impact evidence	15% of landing zone
Data confidence	**0.41**

NEAREST DESTINATION PER TIER — LEAST-COST ROUTE

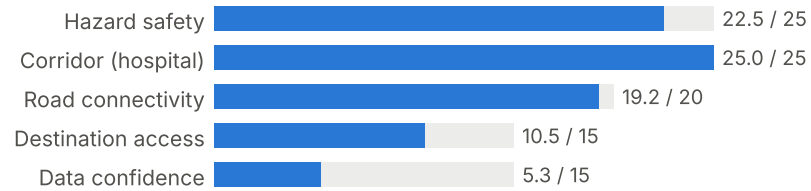
Tier	Destination	km	on road
Hospital	Mu'a Hospital	**5.5**	98%
School / clinic	Unnamed school	3.5	97%
Community / worship	Unnamed place of worship	3.1	97%

Hospital route: max slope 11°, mean impact excess 0.08, impact evidence 67% of route.

ROBUSTNESS

Rank range	**7–12 across 12 scenarios**
In top 5	0 of 12 scenarios

SCORE BREAKDOWN (POINTS)



NOT ASSESSED — VERIFY BEFORE USE

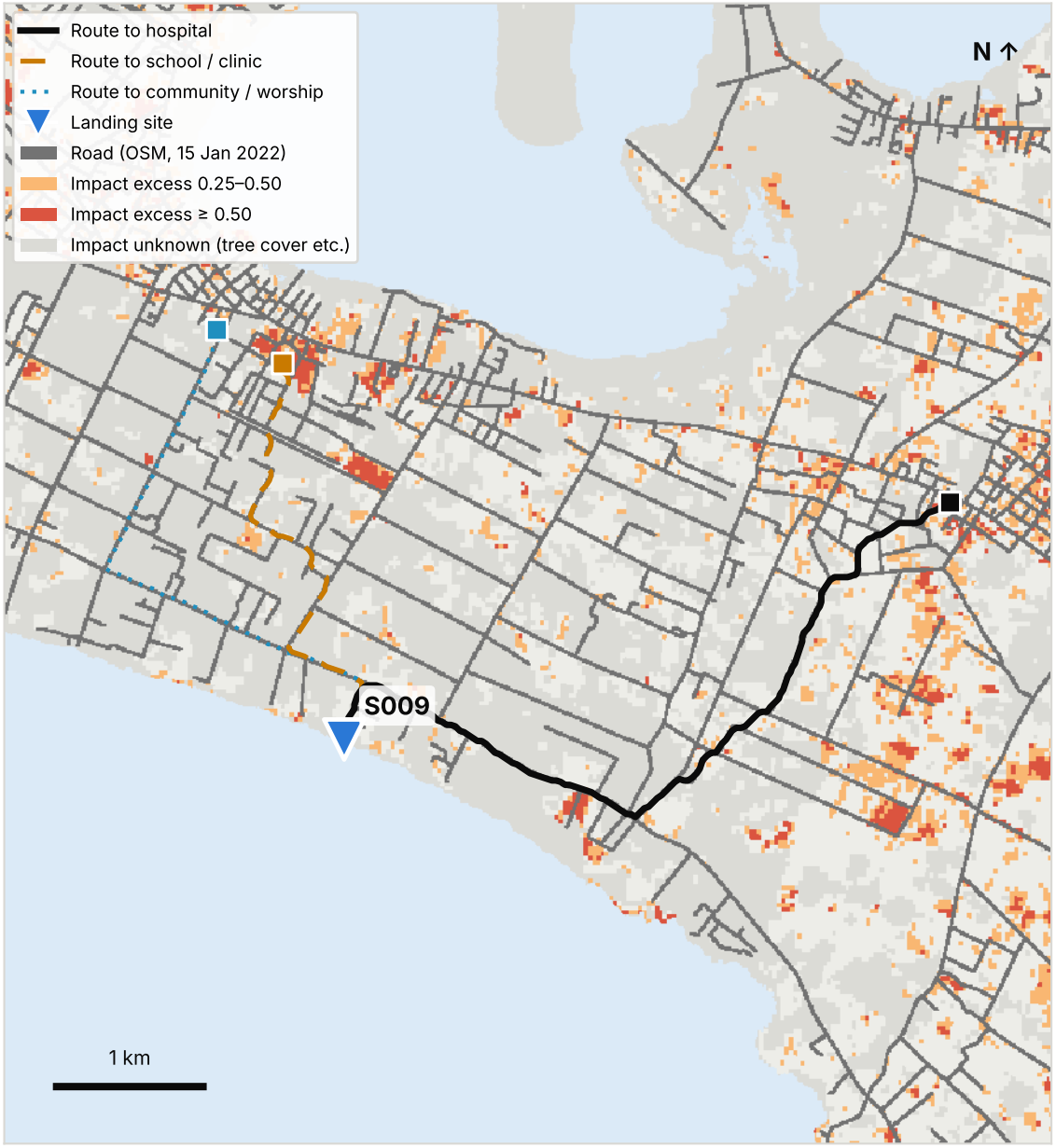
Water depth, approach and reef; beach gradient, surf and tide; obstacles; ground bearing capacity; current road and bridge condition.
Under half of landing zone + route has impact evidence (mostly tree cover).

Shore-Gate: Egress · screening prototype (GEOG761, University of Auckland). Data: Sentinel-1/2, ALOS-2 DPM, ESA WorldCover 2021, OSM (15 Jan 2022), Copernicus EMSR558. Score = weighted sum of five normalised criteria.
Decision support only. Results come from open satellite data and models; they must be verified by aerial reconnaissance and ground survey before operational use.

Landing site S009

Rank 9 of 112 · score 82.5 / 100 · Tongatapu, post-eruption (Jan 2022)

RECONNAISSANCE PRIORITY: HIGH
top-ranked, weak evidence



LOCATION

Lat / Lon	-21.20684°, -175.22216°
UTM zone 1S	E 684535 N 7653925
Lagoon mouth	7.7 km

LANDING ZONE (250 M)

Road access	32 m to road (unsealed)
Impact excess	0.04 (0 = typical, 1 = max)
Impact evidence	43% of landing zone
Data confidence	**0.46**

NEAREST DESTINATION PER TIER — LEAST-COST ROUTE

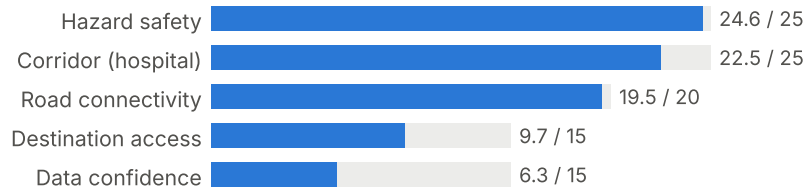
Tier	Destination	km	on road
Hospital	Vainī Hospital	**6.1**	94%
School / clinic	Tonga College 'Atele	3.6	95%
Community / worship	Unnamed place of worship	4.5	98%

Hospital route: max slope 13°, mean impact excess 0.09, impact evidence 49% of route.

ROBUSTNESS

Rank range	**6–14 across 12 scenarios**
In top 5	0 of 12 scenarios

SCORE BREAKDOWN (POINTS)



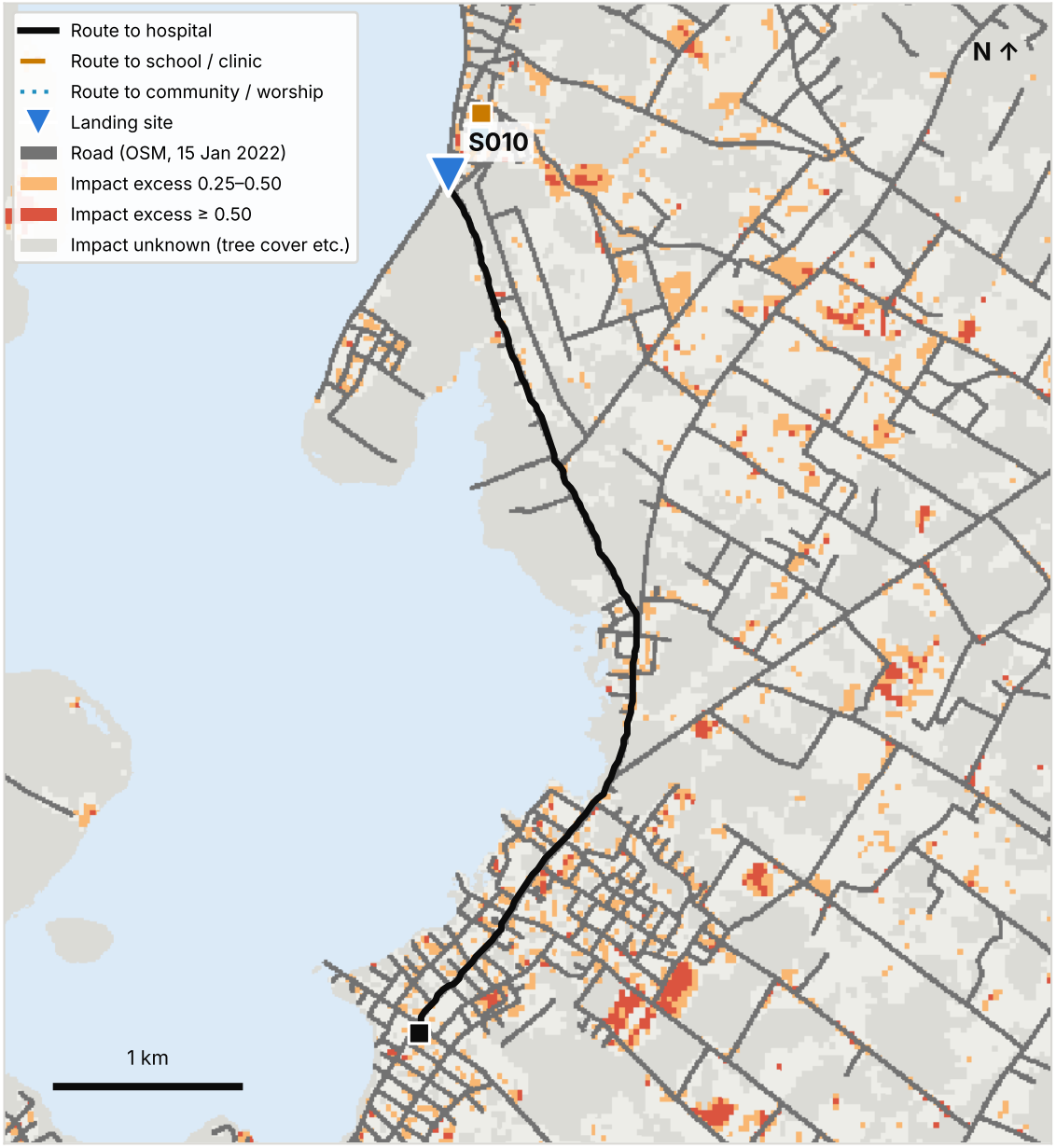
NOT ASSESSED — VERIFY BEFORE USE

Water depth, approach and reef; beach gradient, surf and tide; obstacles; ground bearing capacity; current road and bridge condition.
Under half of landing zone + route has impact evidence (mostly tree cover).

Shore-Gate: Egress · screening prototype (GEOG761, University of Auckland). Data: Sentinel-1/2, ALOS-2 DPM, ESA WorldCover 2021, OSM (15 Jan 2022), Copernicus EMSR558. Score = weighted sum of five normalised criteria.
Decision support only. Results come from open satellite data and models; they must be verified by aerial reconnaissance and ground survey before operational use.

Landing site S010

Rank 10 of 112 · score 82.3 / 100 · Tongatapu, post-eruption (Jan 2022)



LOCATION

Lat / Lon	-21.14253°, -175.12194°
UTM zone 1S	E 695025 N 7660925
Lagoon mouth	1.2 km

LANDING ZONE (250 M)

Road access	10 m to road (sealed)
Impact excess	0.21 (0 = typical, 1 = max)
Impact evidence	38% of landing zone
Data confidence	**0.52**

NEAREST DESTINATION PER TIER — LEAST-COST ROUTE

Tier	Destination	km	on road
Hospital	Mu'a Hospital	**5.7**	99%
School / clinic	Unnamed school	0.5	91%
Community / worship	Unnamed place of worship	0.4	94%

Hospital route: max slope 13°, mean impact excess 0.10, impact evidence 65% of route.

ROBUSTNESS

Rank range	**8–10 across 12 scenarios**
In top 5	0 of 12 scenarios

SCORE BREAKDOWN (POINTS)



NOT ASSESSED — VERIFY BEFORE USE

Water depth, approach and reef; beach gradient, surf and tide; obstacles; ground bearing capacity; current road and bridge condition.

Shore-Gate: Egress · screening prototype (GEOG761, University of Auckland). Data: Sentinel-1/2, ALOS-2 DPM, ESA WorldCover 2021, OSM (15 Jan 2022), Copernicus EMSR558. Score = weighted sum of five normalised criteria.
Decision support only. Results come from open satellite data and models; they must be verified by aerial reconnaissance and ground survey before operational use.